GRACEDAVE J. TORSINO

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Portfolio

PROFESSIONAL SUMMARY

Computer Science graduate (Cum Laude) with experience in **software development, robotics, artificial intelligence**, and **experimental research**. Experienced in developing software solutions, **data collection**, **analysis**, and evaluation pipelines, conducting experimental research, and solving technical problems through programming and systematic analysis.

SKILLS

Programming / Database: C, Python, JavaScript, TypeScript, PostgreSQL, SQL
Data Analysis: Google Sheets, Python, Quantitative Analysis, Data Visualization, Data Collection, Dataset Processing, Experimental Evaluation
Development Tools: Git, GitHub, Conda
Machine Learning & Computer Vision: PyTorch, Diffusion Policy, OpenCV, YOLOv5
Robotics & Simulation: ROS2, Webots, Teleoperation Systems
Operating Systems / Productivity: Windows, Linux (Ubuntu), Google Workspace, Microsoft Office

EXPERIENCE

AI and Robotics Laboratory (AIRLab) at Caraga State University – Main Campus — *Intern*
Project: Teaching RoArm-M2 Push and Pick-and-Place Behaviors
March 2026 – May 2026

- Collected and processed teleoperation demonstrations for robotic pushing and pick-and-place tasks
- Evaluated policy performance using task success rates, failure counts, and completion times
- Adapted an existing vision-based Diffusion Policy and developed a leader–follower teleoperation workflow
- Conducted sim-to-real experiments to assess policy performance on physical robotic hardware

PROJECTS

6dMetrix — Sales Reporting and Inventory Dashboard

- Developed a system summarizing net sales, average transaction value, top-selling products, and sales by category across selectable date ranges
- Implemented CSV bulk import with row-level validation, duplicate reference detection, category-conflict checks, and atomic imports to maintain data integrity
- Built inventory tracking features using stock movements and completed sales, including low-stock and out-of-stock alerts
- Generated downloadable PDF and CSV reports and created interactive charts for exploring sales trends using a synthetic motorcycle-dealership dataset

Prototype 0 (<p.0>): Custom Imperative Programming Language

- Designed and implemented a custom imperative programming language and its C-based compiler
- Designed the application to run offline as a web app or Android app, with backup and restore functionality

Individual Undergraduate Thesis: Sim-to-Sim Transfer of Low-Dimensional Diffusion Policy for SO-ARM100 Pick-and-Place Using Webots Demonstrations

- Conducted training and evaluation of a low-dimensional Diffusion Policy for SO-ARM100 pick-and-place tasks in Webots using demonstrations collected through real-to-sim teleoperation

- Developed teleoperation-based data collection and evaluation pipelines for sim-based Diffusion Policy experiments
- Adapted and integrated an existing Diffusion Policy implementation into a custom Webots workflow via dataset, task, and workspace configuration
- Designed and implemented environment randomization for data collection, policy evaluation, and generalization testing in Webots

a-Nventory — Inventory Management System Research and Design

- Designed a questionnaire for faculty, technical team members, and staff to investigate inventory-tracking practices, record-keeping challenges, perceived need for automation, and desired system features

- Interviewed the technical-team and student assistants to document existing borrowing and return workflows, inventory discrepancies, and equipment-tracking challenges
- Identified recurring issues, including missing equipment components, unlogged transactions, data loss risks, and difficulty tracing overdue or misplaced items
- Translated stakeholder findings into proposed system requirements, including QR-code scanning, equipment reservations, transaction tracking, and automated notifications
- Created system diagrams, use-case models, and UI wireframes to represent proposed workflows and functionality, both for admins and non-admins

6d CIMS — Customizable Inventory Management System

- Developed a customizable inventory management system that allows organizations to define their own inventory tables and fields
- Implemented tool borrowing and partial-return tracking, consumed-material records with supporting attachments, and low-stock and overdue alerts
- Designed for offline use as a web application or Android app, with backup and restore functionality

StateFlow: Automata Simulator

- Developed a web-based application that converts natural-language descriptions into finite automata, visualizes state transitions, and simulates user-provided input strings through an AI-powered backend

Real-Time Face Anonymization (Python + OpenCV)

- Developed a real-time face anonymization system using OpenCV for privacy-preserving video processing

3D Projectile Simulator

- Developed a physics-based 3D projectile motion simulator with interactive visualization

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EDUCATION

Caraga State University – Main Campus — Bachelor of Science in Computer Science (2022 – 2026)
Honors: Cum Laude

Mindanao State University – Iligan Institute of Technology — Bachelor of Science in Physics
Completed five (5) semesters of the program

PRESENTATION

Students' Congress on Innovation, Informatics, and Computing (SCII-Com) 2026
Research Presenter — *May 22, 2026*